

Supplementary Material for “Adaptive Estimation Distribution Multi-task Optimization”

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S.I. THEORETICAL ANALYSIS OF AEDMTO

In this section, the impact of the AED strategy on the task distribution and the convergence proof of AEDMTO are analyzed. Herein, we only analyze the MTOP with two tasks since the analysis is easy to extend to many-task scenarios. During the KT process, the target task t may meet two cases.

In the first case, with probability $P(\text{Case 1}) = 1 - f_{\text{KT}}$, KT is not performed on the target task t . In this case, the current distribution of the target task t is maintained:

$$p(X | \text{Case 1}) = p_t(X; \mu_t, \sigma_t) \quad (\text{S1})$$

where μ_t and σ_t stand for the centroid and the standard deviation of the population for the target task t , $p_t(X; \mu_t, \sigma_t)$ is the probability density function of the target task (abbreviated as $p_t(X)$ hereafter).

In the second case, KT is performed with probability $P(\text{Case 2}) = f_{\text{KT}}$. In this case, the final population is the mixture distribution of the parent population and the offspring population (containing the distribution information from the source task s), as:

$$p(X | \text{Case 2}) = 0.5 \times p_t(X) + 0.5 \times p_t^c(X) \quad (\text{S2})$$

where $p_t^c(X)$ is the distribution of the offspring. Herein, we mainly analyze the distribution $p_t^c(X)$.

First, when selecting individuals, the roulette wheel selection (RWS) is employed, which requires the normalized selection probabilities for items. The probabilities for individual selection *ins* in AED strategy are calculated by the Gaussian distribution-based probability density function of the target task ($p_t(X)$). Notably, the integral of the Gaussian distribution equals 1.

$$\int_X p_t(X) dX = 1 \quad (\text{S3})$$

Hence, $p_t(X)$ can be used as the final selection probabilities for RWS. Based above, the information provided by the source task possesses a distribution $C_1 \times p_t(X) \times p_s(X; \mu_s, \sigma_s)$, where μ_s and σ_s stand for the centroid and the standard deviation of the population for the source task s , $p_s(X; \mu_s, \sigma_s)$ is the probability density function of the source task (abbreviated as $p_s(X)$ hereafter), C_1 is a correction coefficient to make the integral of the probability density function equal to 1, as:

$$C_1 = 1 / \int_X p_t(X) \times p_s(X) dX \quad (\text{S4})$$

Then, the final population of the target task t after KT consists of both parents and the offspring with the same numbers. In AED strategy, the offspring is generated by the binomial crossover operator, and the information from the source task flows into the target task with a rate CR (whose expectation $E(CR) = 0.5$). Hence, the offspring population $p_t^c(X)$ is:

$$\begin{aligned} p_t^c(X) &= 1/2 \times p_t(X) + 1/2 \times [(1 - E(CR)) \times p_t(X) + E(CR) \times C_1 \times p_t(X) \times p_s(X)] \\ &= 3/4 \times p_t(X) + 1/4 \times C_1 \times p_t(X) \times p_s(X) \end{aligned} \quad (\text{S5})$$

In all, combining the two cases, the expected population of the target task in the next generation can be calculated as:

$$\begin{aligned} p^{\text{AED}}(X) &= P(\text{Case 1}) \times p(X | \text{Case 1}) + P(\text{Case 2}) \times p(X | \text{Case 2}) \\ &= (1 - f_{\text{KT}} / 4) \times p_t(X) + f_{\text{KT}} / 4 \times C_1 \times p_t(X) \times p_s(X) \end{aligned} \quad (\text{S6})$$

In fact, according to the proof in MFEA-II [S1], the convergence of AEDMTO can be proved. In MFEA-II, the population for the t^{th} task in the next generation can be described as the mixture distribution, defined as:

$$p_t^{\text{Next}}(X) = \alpha_t \times p_t(X) + \sum_{s \neq t} \alpha_s \times p_s(X) \quad (\text{S7})$$

where α_t and α_s are the mixture coefficients for each population, $\alpha_t + \sum_{s \neq t} \alpha_s = 1$, and each α_t or $\alpha_s \geq 0$. If given $p'(X)$ is positive and continuous in the search space for all tasks, an EMTO algorithm converges to the global optimum for each task if:

1) AEDMTO follows the (μ, λ) selection strategy.

Proof: In AEDMTO, elitist selection is employed in each generation.

2) $\alpha_t > \mu / \lambda$, where μ denotes the number of parents, and λ denotes the number of both parents and offspring.

Proof: In AEDMTO, $\mu = N$, $\lambda = 2 \times N$. Hence, $\alpha_t = 1 - f_{\text{KT}} / 4 > \mu / \lambda = 0.5$.

Therefore, each task in AEDMTO can converge to the global optimum.

Next, we continue to analyze the impact of AED strategy on the population distribution. In AED strategy, the KT frequency

f_{KT} is defined as the weighted average of the individual similarity *ins*. Note that the weights are determined by the individual similarity *ins* of individuals, which are consistent with the values of $p_t(X)$. Therefore, $p_t(X)$ can be used as the weights when calculating f_{KT} . Based above, the KT frequency f_{KT} can be calculated as:

$$\begin{aligned} f_{KT} &= C_2 \times \int_X p_t(X) \times p_t(X) \times p_s(X) dX \\ C_2 &= 1 / \int_X [p_t(X)]^3 dX \end{aligned} \quad (S8)$$

where C_2 is a correction coefficient to guarantee $f_{KT} = 1$ when $p_s(X) \equiv p_t(X)$.

Meanwhile, in a variant using random selection of individuals and fixed KT frequency, the expected distribution of the target task in the next generation can be calculated as:

$$p^{\text{RAND}}(X) = (1 - f_{KT} / 4) \times p_t(X) + f_{KT} / 4 \times p_s(X) \quad (S9)$$

Inner product is a commonly used similarity metric for two distributions, which can be used here to evaluate the similarity between the parent population of the target task and the expected population for the next generation. The inner products between $p_t(X)$ and $p^{\text{AED}}(X)$, and between $p_t(X)$ and $p^{\text{RAND}}(X)$, respectively, are calculated as:

$$\begin{aligned} I^{\text{AED}} &= \int_X [(1 - f_{KT} / 4) \times p_t(X) + f_{KT} / 4 \times C_1 \times p_t(X) \times p_s(X)] \times p_t(X) dX \\ &= (1 - f_{KT} / 4) \int_X [p_t(X)]^2 dX + f_{KT} / 4 \times \int_X [p_t(X)]^2 p_s(X) dX / \int_X p_t(X) p_s(X) dX \end{aligned} \quad (S10)$$

$$\begin{aligned} I^{\text{RAND}} &= \int_X [(1 - f_{KT} / 4) \times p_t(X) + f_{KT} / 4 \times p_s(X)] \times p_t(X) dX \\ &= (1 - f_{KT} / 4) \int_X [p_t(X)]^2 dX + f_{KT} / 4 \times \int_X p_t(X) p_s(X) dX \end{aligned} \quad (S11)$$

With Cauchy-Schwarz inequality, we have:

$$\begin{aligned} \left[\int_X p_t(X) p_s(X) dX \right]^2 &= \left[\int_X p_t(X) \times 1 \times p_s(X) dX \right]^2 \\ &\leq \int_X [p_t(X)]^2 p_s(X) dX \times \int_X 1^2 \times p_s(X) dX \\ &= \int_X [p_t(X)]^2 p_s(X) dX \end{aligned} \quad (S12)$$

Therefore, under the same KT frequency f_{KT} , $I^{\text{AED}} \geq I^{\text{RAND}}$, i.e., with the adaptive selection of individuals, the expected population of the target task in the next generation $p^{\text{AED}}(X)$ has higher similarity with the original population of the target task $p_t(X)$ than $p^{\text{RAND}}(X)$, showing the effectiveness of the individual selection mechanism.

We continue to analyze the impact of the AED strategy on KT frequency. Note that when $p_s(X) \approx p_t(X)$, the inner product between $p_t(X)$ and $p_t(X)$ is a high value, while f_{KT} (which is also an inner product form in Eq. (S8)) also has a high value, which further enlarges the similarity between $p_t(X)$ and $p^{\text{AED}}(X)$. On the contrary, when $p_s(X)$ is dissimilar to $p_t(X)$, f_{KT} can decrease to a lower value, which filters the unrelated information in the source task and avoids negative transfer.

Based on the theoretical analysis above, the distribution difference between the expected population $p^{\text{AED}}(X)$ and the target population $p_t(X)$ can be reduced with the similarity metrics (from both the individual and population aspects) in AED strategy, maintaining high-quality of the information in target task. Meanwhile, AED adaptively adjusts f_{KT} according to the task similarity, allowing more positive knowledge to be transferred between similar tasks and forbidding negative KT between dissimilar tasks.

S.II. EXPERIMENTAL RESULTS WITH CONFIDENCE INTERVALS AND STATISTICAL POWERS

In this section, the complete experimental results of AEDMTO with confidence intervals and statistical powers on CEC2017, CEC2022, and SCP application are revealed.

A. Confidence Interval

The confidence intervals are the estimated ranges $[lb, hb]$ within which the results fall. Suppose the sample follows a Gaussian distribution $N(\mu_0, (\sigma_0)^2)$, the confidence interval of the sample is shown as:

$$1 - \alpha = P\{lb < Y \leq hb\}, Y \sim N(\mu_0, (\sigma_0)^2) \quad (S13)$$

$$lb = z_{\alpha/2}, hb = z_{1-\alpha/2} \quad (S14)$$

where z_p is the critical value corresponding to cumulative probability p of the Gaussian distribution $N(\mu_0, (\sigma_0)^2)$. Herein, the confidence interval at 95% confidence level ($\alpha = 0.05$) for each result across the 30 runs can be computed.

B. Statistical Power

The statistical power is the probability of avoiding Type-II error in a hypothesis test, i.e., when the null hypothesis is false, the probability that the samples can lead to rejection of the null hypothesis. Herein, for two algorithms A and B on one task, the null hypothesis is that the results obtained by A and B are similar. Suppose the results obtained by A and B both follow a Gaussian distribution. The null hypothesis is: $\mu_1 = \mu_0$, and the alternative hypothesis is: $\mu_1 \neq \mu_0$, where μ_0 and μ_1 are the mean values of the 30 runs obtained by A and B, respectively. The statistical power is defined as:

$$power = 1 - \beta = P\{Y \leq z_{0.025} \cup Y > z_{0.975}\}, Y \sim N(\mu_1, (\sigma_1)^2) \quad (S15)$$

$$P\{X \leq z_p\} = p, X \sim N(\mu_0, (\sigma_0)^2) \quad (S16)$$

where β is the probability of type-II error, z_p is the critical value corresponding to cumulative probability p of the Gaussian distribution $N(\mu_0, (\sigma_0)^2)$, and σ_0 and σ_1 are the standard deviations of the results obtained by A and B, respectively. Higher statistical power usually indicates more obvious difference between the results. In practice, if two samples have a high statistical power greater than 0.8 [S2], the two samples can be considered significantly different.

C. Complete Experimental Results on CEC2017, CEC2022, and SCP

TABLE S.1
COMPLETE EXPERIMENTAL RESULTS OF AEDMTO AND COMPARED ALGORITHMS ON CEC2017

CEC2017		AEDMTO	MFEA	MFEA-II	MFDE	MFEA-DGD	RLMFEA	MFEA-FL	MTGA	AEMTO	MKTDE	OTMTO	BLKT-DE	
CIHS	T1	mean	0.00e+00	1.58e-01(+)	4.29e-02(+)	7.88e-04(+)	0.00e+00(=)	2.48e-02(+)	1.17e-03(+)	2.24e-02(+)	1.03e-01(+)	2.54e-05(+)	1.19e-02(+)	4.92e-02(+)
		interval	[0.00e+00, 0.00e+00]	[4.79e-02, 2.69e-01]	[1.49e-02, 7.08e-02]	[3.71e-03, 5.28e-03]	[0.00e+00, 0.00e+00]	[1.88e-02, 6.85e-02]	[7.26e-03, 9.60e-03]	[1.07e-02, 5.54e-02]	[5.51e-03, 2.01e-01]	[2.35e-04, 2.86e-04]	[6.31e-03, 3.01e-02]	[3.38e-02, 1.32e-01]
		power	-	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	T2	mean	5.33e-14	1.25e+02(+)	2.59e+01(+)	1.55e+00(+)	1.74e-08(+)	5.22e+01(+)	6.78e-01(+)	7.19e+01(+)	1.55e+02(+)	5.83e-02(+)	1.35e+01(+)	1.07e+02(+)
		interval	[1.84e-14, 8.83e-14]	[4.70e+01, 2.03e+02]	[1.85e+00, 5.37e+01]	[7.62e+00, 1.07e+01]	[6.03e-09, 4.08e-08]	[1.89e+01, 1.23e+02]	[4.01e+00, 5.37e+00]	[3.29e+01, 1.11e+02]	[2.34e+01, 2.87e+02]	[5.45e-01, 6.62e-01]	[1.95e+01, 4.65e+01]	[1.42e+01, 2.00e+02]
		power	-	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
CIMS	T1	mean	4.40e-10	3.30e+00(+)	9.85e-01(+)	1.27e-02(+)	3.49e-08(+)	2.21e+00(+)	4.89e-03(+)	1.49e+00(+)	1.06e+00(+)	1.60e-08(+)	2.09e-03(+)	3.02e-01(+)
		interval	[2.61e-10, 6.19e-10]	[2.09e+00, 4.52e+00]	[8.93e-02, 1.88e+00]	[7.34e-02, 9.88e-02]	[7.33e-08, 1.43e-07]	[1.28e+00, 3.14e+00]	[1.44e-03, 1.12e-02]	[2.90e-01, 2.68e+00]	[1.50e-04, 2.13e+00]	[1.82e-08, 5.01e-08]	[9.06e-05, 4.09e-03]	[2.49e-02, 6.30e-01]
		power	-	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	1.00
	T2	mean	0.00e+00	1.65e+02(+)	2.41e+01(+)	2.75e-01(+)	1.98e-08(+)	7.97e+01(+)	1.99e-02(+)	7.01e+01(+)	1.25e+02(+)	2.75e-13(=)	2.76e-03(+)	1.07e+02(+)
		interval	[0.00e+00, 0.00e+00]	[6.86e+01, 2.61e+02]	[2.06e+00, 5.02e+01]	[1.73e+00, 2.28e+00]	[1.19e-08, 5.16e-08]	[2.88e+01, 1.31e+02]	[3.86e-02, 7.85e-02]	[2.41e+01, 1.16e+02]	[2.64e+01, 2.75e+02]	[1.30e-12, 1.85e-12]	[2.13e-03, 7.66e-03]	[3.34e+01, 1.81e+02]
		power	-	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
CILS	T1	mean	2.08e+01	2.04e+01(-)	2.12e+01(+)	2.12e+01(+)	5.85e+00(-)	2.00e+01(-)	1.88e-02(-)	2.12e+01(+)	2.12e+01(+)	2.12e+01(+)	2.12e+01(+)	2.12e+01(+)
		interval	[2.06e+01, 2.10e+01]	[2.02e+01, 2.06e+01]	[2.11e+01, 2.13e+01]	[2.11e+01, 2.13e+01]	[8.44e+00, 2.01e+01]	[1.99e+01, 2.01e+01]	[4.67e-04, 3.71e-02]	[2.09e+01, 2.14e+01]	[2.11e+01, 2.13e+01]	[2.08e+01, 2.15e+01]	[2.12e+01, 2.13e+01]	[2.11e+01, 2.13e+01]
		power	-	0.99	1.00	1.00	1.00	1.00	1.00	0.83	1.00	0.78	1.00	1.00
	T2	mean	2.33e+03	2.43e+03(=)	1.81e+03(-)	1.20e+04(+)	2.03e+03(-)	3.29e+03(+)	1.40e-02(-)	5.28e+03(+)	1.04e+04(+)	1.20e+04(+)	8.32e+03(+)	6.36e+03(+)
		interval	[1.07e+03, 3.60e+03]	[1.47e+03, 3.38e+03]	[1.07e+03, 2.54e+03]	[8.45e+03, 1.56e+04]	[5.79e+03, 9.85e+03]	[2.47e+03, 4.12e+03]	[1.20e-02, 4.00e-02]	[3.95e+03, 6.62e+03]	[9.18e+03, 1.15e+04]	[9.00e+03, 1.49e+04]	[7.55e+03, 9.09e+03]	[2.71e+03, 1.00e+04]
		power	-	0.01	0.02	1.00	0.75	0.24	1.00	0.99	1.00	1.00	1.00	0.93
PIHS	T1	mean	2.07e+02	4.67e+02(+)	1.33e+02(-)	8.94e+01(-)	0.00e+00(-)	2.36e+02(+)	8.91e+01(-)	7.22e+01(-)	4.44e+02(+)	3.65e+02(+)	4.59e-05(-)	1.67e+02(-)
		interval	[4.26e+01, 3.72e+02]	[2.66e+02, 6.68e+02]	[8.73e+01, 1.79e+02]	[5.55e+01, 1.23e+02]	[0.00e+00, 0.00e+00]	[1.50e+02, 3.21e+02]	[3.43e+01, 1.44e+02]	[2.24e+01, 1.22e+02]	[4.12e+02, 4.76e+02]	[3.27e+02, 4.03e+02]	[4.57e-05, 1.38e-04]	[2.51e+01, 3.58e+02]
		power	-	0.82	0.00	0.00	1.00	0.00	0.05	0.12	1.00	0.36	1.00	0.12
	T2	mean	7.79e-02	2.03e+00(+)	1.92e-02(+)	1.30e-04(+)	4.16e+01(+)	3.63e-02(+)	2.37e-02(+)	8.90e-03(+)	3.34e+00(+)	3.39e-01(+)	9.89e-07(-)	2.82e-01(+)
		interval	[-6.68e-01, 8.24e-01]	[-5.18e-01, 4.59e+00]	[-2.07e-02, 5.92e-02]	[-1.69e-04, 4.29e-04]	[2.84e+00, 8.03e+01]	[-1.97e-02, 9.47e-01]	[-1.11e-04, 4.75e-02]	[-1.96e-02, 3.74e-02]	[3.86e-01, 6.29e+00]	[1.25e-01, 5.53e-01]	[-1.08e-06, 3.05e-06]	[-1.31e-01, 6.95e-01]
		power	-	0.84	0.00	0.00	1.00	0.00	0.00	0.00	0.96	0.00	0.00	0.01
PIMS	T1	mean	2.74e-03	2.17e+00(+)	4.38e-01(+)	4.26e-03(=)	2.12e+00(+)	1.54e+00(+)	2.35e-01(+)	5.41e-01(+)	1.47e-01(+)	3.94e-03(+)	4.38e-03(+)	4.02e-01(+)
		interval	[-3.61e-03, 9.08e-03]	[8.91e-01, 3.44e+00]	[-9.67e-01, 1.84e+00]	[-6.17e-03, 1.47e-02]	[1.48e+00, 2.77e+00]	[4.63e+01, 2.61e+00]	[-3.11e-01, 7.80e-01]	[-6.64e-01, 1.75e+00]	[7.72e-02, 2.16e-01]	[1.23e-03, 6.65e-03]	[4.08e-04, 8.34e-03]	[-2.71e-01, 1.08e+00]
		power	-	1.00	0.99	0.25	1.00	1.00	0.99	0.99	1.00	0.00	0.01	0.99
	T2	mean	4.99e+01	2.43e+02(+)	1.49e+02(+)	7.91e+01(+)	4.41e+00(-)	1.60e+02(+)	4.77e+01(+)	2.45e+02(+)	9.53e+01(+)	7.49e+01(+)	1.15e+02(+)	7.04e+01(+)
		interval	[2.77e+01, 7.22e+01]	[4.18e+01, 4.43e+02]	[-1.53e+02, 4.50e+02]	[3.73e+01, 1.21e+02]	[-6.11e+00, 1.49e+01]	[1.73e+01, 3.03e+02]	[4.68e+01, 4.87e+01]	[-4.01e+02, 8.91e+02]	[9.06e+01, 9.99e+01]	[3.78e+01, 1.12e+02]	[1.04e+01, 2.20e+02]	[1.26e+01, 1.28e+02]
		power	-	0.97	0.91	0.64	1.00	0.92	0.00	0.95	1.00	0.56	0.84	0.55
PILS	T1	mean	0.00e+00	1.87e+01(+)	1.63e+00(+)	5.57e-01(+)	3.94e-09(+)	2.90e+00(+)	1.74e-02(+)	2.35e+00(+)	1.24e+01(+)	4.46e+00(+)	3.95e-02(+)	6.33e+00(+)
		interval	[0.00e+00, 0.00e+00]	[1.01e+01, 2.73e+01]	[7.39e-01, 2.52e+00]	[-5.03e-01, 1.62e+00]	[-1.32e-08, 2.11e-08]	[1.67e+00, 4.14e+00]	[7.60e-03, 2.72e-02]	[1.18e+00, 3.53e+00]	[1.02e+00, 2.39e+01]	[5.35e-01, 8.39e+00]	[-3.67e-01, 4.46e-01]	[-8.60e+00, 2.13e+01]
		power	-	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	T2	mean	0.00e+00	1.98e+01(+)	5.87e+00(+)	9.86e-02(+)	1.46e-03(+)	3.59e+00(+)	1.04e-01(+)	3.57e+00(+)	9.78e+00(+)	1.45e+00(+)	6.00e-02(+)	3.48e+00(+)
		interval	[0.00e+00, 0.00e+00]	[8.35e+00, 3.12e+01]	[9.35e-01, 1.08e+01]	[-2.05e-01, 4.02e-01]	[9.00e-04, 2.03e-03]	[1.05e+00, 6.14e+00]	[4.08e-02, 1.67e-01]	[-8.96e-01, 8.03e+00]	[-9.44e-01, 2.05e+01]	[-8.36e-01, 3.73e+00]	[-3.47e-01, 4.67e-01]	[-2.61e+00, 9.56e+00]
		power	-	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
NIHS	T1	mean	4.79e+01	4.39e+02(+)	2.00e+02(+)	9.17e+01(+)	2.90e+00(-)	2.06e+02(+)	4.79e+01(-)	2.12e+02(+)	3.54e+02(+)	4.67e+01(-)	9.34e+01(+)	1.35e+02(+)
		interval	[4.76e+01, 4.83e+01]	[3.95e+01, 8.38e+02]	[1.06e+02, 2.94e+02]	[1.86e+01, 1.65e+02]	[-5.00e+00, 1.08e+01]	[8.62e+01, 3.25e+02]	[4.76e+01, 4.81e+01]	[-4.21e+02, 8.44e+02]	[5.31e+01, 6.56e+02]	[4.47e+01, 4.87e+01]	[-2.25e+01, 2.09e+02]	[-1.52e+02, 4.23e+02]
		power	-	1.00	1.00	1.00	1.00	1.00	0.01	1.00	1.00	0.87	1.00	1.00
	T2	mean	2.09e-04	2.07e+02(+)	7.32e+01(+)	2.44e+01(+)	0.00e+00(-)	1.12e+02(+)	3.74e-02(+)	7.69e+01(+)	2.92e+02(+)	2.03e+00(+)	1.30e+02(+)	1.51e+02(+)
		interval	[-4.04e-04, 8.22e-04]	[8.94e+01, 3.25e+02]	[-3.75e+01, 1.84e+02]	[-3.39e+00, 5.22e+01]	[0.00e+00, 0.00e+00]	[3.80e+01, 1.87e+02]	[-3.14e-02, 1.06e-01]	[3.94e+01, 1.14e+02]	[2.16e+02, 3.68e+02]	[-7.11e+00, 1.12e+01]	[-1.70e+02, 4.29e+02]	[4.60e+01, 2.57e+02]
		power	-	1.00	1.00	1.00	0.00	1.00	0.99	1.00	1.00	1.00	1.00	1.00
NIMS	T1	mean	2.97e-03	2.06e-01(+)	2.62e-02(+)	2.48e-03(=)	1.10e-01(+)	7.42e-02(+)	1.33e-02(+)	1.26e-02(+)	2.79e-01(+)	7.84e-03(+)	1.44e-03(+)	9.21e-02(+)
		interval	[-8.42e-03, 1.44e-02]	[6.32e-02, 3.48e-01]	[2.30e-03, 5.00e-02]	[-9.19e-03, 1.41e-02]	[-9.71e-03, 2.30e-01]	[1.79e-02, 1.30e-01]	[3.07e-03, 2.35e-02]	[-9.14e-03, 3.44e-02]	[1.60e-01, 3.99e-01]	[2.98e-03, 1.27e-02]	[-4.98e-03, 7.87e-03]	[-1.27e-02, 1.97e-01]
		power	-	1.00	0.84	0.06	0.97	0.98	0.42	1.00	1.00	0.00	0.00	0.96
	T2	mean	3.19e+00	2.65e+01(+)	1.59e+01(+)	3.51e+00(+)	7.52e-03(-)	1.87e+01(+)	8.26e-01(-)	1.01e+01(+)	8.59e+00(+)	1.56e+00(-)	5.97e+00(+)	3.21e+00(+)
		interval	[9.55e-01, 5.43e+00]	[1.94e+01, 3.36e+01]	[7.73e+00, 2.41e+01]	[1.27e+00, 5.76e+00]	[2.50e-03, 1.25e-02]	[1.09e+01, 2.65e+01]	[2.64e-01, 1.39e+00]	[1.19e+00, 1.91e+01]	[6.92e+00, 1.03e+01]	[9.21e-01, 2.20e+00]	[2.28e+00, 9.66e+00]	[1.91e+00, 4.50e+00]
		power	-	1.00	0.99	0.06	1.00	1.00	0.67	0.87	1.00	0.03	0.62	0.00
NILS	T1	mean	1.68e+02	5.50e+02(+)	1.80e+02(+)	1.65e+02(+)	0.00e+00(-)	3.35e+02(+)	1.08e+02(-)	2.69e+02(+)	4.55e+02(+)	4.27e+02(+)	4.60e+02(+)	3.79e+02(+)
		interval	[3.03e+01, 3.07e+02]	[2.73e+02, 8.26e+02]	[7.74e+00, 3.52e+02]	[-4.32e+01, 3.72e+02]	[0.00e+00, 0.00e+00]	[1.49e+02, 5.22e+02]	[-1.59e+01, 2.33e+02]	[1.46e+02, 3.93e+02]	[4.21e+02, 4.90e+02]	[3.87e+02, 4.68e+02]	[4.17e+02, 5.02e+02]	[2.30e+02, 5.28e+02]
		power	-	0.96	0.12	0.19	1.00	0.62	0.11	0.28	1.00	1.00	1.00	0.83
	T2	mean	2.93e+03	2.41e+03(-)	1.69e+03(-)	4.24e+03(+)	9.06e+03(+)	3.26e+03(+)	4.85e+03(+)	4.43e+03(+)	1.48e+04(+)	2.09e+03(-)	9.82e+03(+)	1.88e+03(-)
		interval	[1.80e+03, 4.06e+03]	[1.40e+03, 3.42e+03]	[9.01e+02, 2.47e+03]	[2.50e+03, 5.98e+03]	[6.73e+03, 1.14e+04]	[2.28e+03, 4.25e+03]	[2.21e+03, 7.49e+03]	[2.96e+03, 5.90e+03]	[1.38e+04, 1.58e+04]	[1.48e+03, 2.69e+03]	[8.56e+03, 1.11e+04]	[-9.95e+02, 4.75e+03]
		power	-	0.12	0.61	0.58	1.00	0.06	0.73	0.69	1.00	0.18	1.00	0.55
Number of "+/=-"		15/1/2	14/1/3	13/4/1	9/1/8	16/1/1	12/1/5	17/0/1	18/0/0	14/1/3	16/0/2	15/1/2		

TABLE S.II
COMPLETE EXPERIMENTAL RESULTS OF AEDMTO AND COMPARED ALGORITHMS ON CEC2022

CEC2022			AEDMTO	MFEA	MFEA-II	MFDE	MFEA-DGD	RLMFEA	MFEA-FL	MTGA	AEMTO	MKTDE	OTMTO	BLKT-DE
Benchmark1	T1	mean	6.03e+02	6.52e+02(+)	6.16e+02(+)	6.06e+02(+)	6.16e+02(+)	6.32e+02(+)	6.03e+02(=)	6.18e+02(+)	6.21e+02(+)	6.09e+02(+)	6.09e+02(+)	6.17e+02(+)
		interval	[6.01e+02, 6.05e+02]	[6.43e+02, 6.62e+02]	[6.08e+02, 6.24e+02]	[5.98e+02, 6.13e+02]	[6.13e+02, 6.19e+02]	[6.18e+02, 6.46e+02]	[6.01e+02, 6.05e+02]	[6.05e+02, 6.30e+02]	[6.06e+02, 6.36e+02]	[6.00e+02, 6.17e+02]	[6.02e+02, 6.16e+02]	[6.07e+02, 6.27e+02]
		power	-	1.00	1.00	0.71	1.00	1.00	0.06	0.98	0.99	0.85	0.91	0.99
	T2	mean	6.03e+02	6.51e+02(+)	6.17e+02(+)	6.06e+02(+)	6.19e+02(+)	6.33e+02(+)	6.03e+02(=)	6.17e+02(+)	6.21e+02(+)	6.08e+02(+)	6.09e+02(+)	6.15e+02(+)
		interval	[6.01e+02, 6.04e+02]	[6.40e+02, 6.62e+02]	[6.05e+02, 6.29e+02]	[5.98e+02, 6.14e+02]	[6.16e+02, 6.22e+02]	[6.17e+02, 6.50e+02]	[6.01e+02, 6.06e+02]	[6.06e+02, 6.29e+02]	[6.05e+02, 6.37e+02]	[6.00e+02, 6.17e+02]	[6.03e+02, 6.16e+02]	[6.04e+02, 6.25e+02]
		power	-	1.00	0.98	0.76	1.00	1.00	0.27	0.99	0.99	0.86	0.94	0.98
Benchmark2	T1	mean	7.00e+02	7.01e+02(+)	7.00e+02(+)	7.00e+02(=)	7.01e+02(+)	7.00e+02(+)	7.00e+02(+)	7.00e+02(+)	7.01e+02(+)	7.00e+02(+)	7.00e+02(=)	7.00e+02(+)
		interval	[7.00e+02, 7.00e+02]	[7.01e+02, 7.01e+02]	[7.00e+02, 7.00e+02]	[7.00e+02, 7.00e+02]	[7.01e+02, 7.01e+02]	[7.00e+02, 7.00e+02]	[7.00e+02, 7.00e+02]	[7.00e+02, 7.00e+02]	[7.01e+02, 7.01e+02]	[7.00e+02, 7.00e+02]	[7.00e+02, 7.00e+02]	[7.00e+02, 7.01e+02]
		power	-	1.00	0.33	0.00	1.00	0.00	0.02	0.45	1.00	0.00	0.00	0.99
	T2	mean	7.00e+02	7.01e+02(+)	7.00e+02(+)	7.00e+02(=)	7.01e+02(+)	7.00e+02(+)	7.00e+02(+)	7.00e+02(+)	7.01e+02(+)	7.00e+02(+)	7.00e+02(+)	7.00e+02(+)
		interval	[7.00e+02, 7.00e+02]	[7.01e+02, 7.01e+02]	[7.00e+02, 7.00e+02]	[7.00e+02, 7.00e+02]	[7.01e+02, 7.01e+02]	[7.00e+02, 7.00e+02]	[7.00e+02, 7.00e+02]	[7.00e+02, 7.00e+02]	[7.01e+02, 7.01e+02]	[7.00e+02, 7.00e+02]	[7.00e+02, 7.00e+02]	[7.00e+02, 7.01e+02]
		power	-	1.00	0.60	0.00	1.00	0.00	0.33	0.62	1.00	0.03	0.00	1.00
Benchmark3	T1	mean	4.13e+05	4.69e+06(+)	2.63e+06(+)	5.75e+06(+)	1.24e+05(-)	1.96e+06(+)	4.67e+05(=)	2.09e+06(+)	2.03e+07(+)	8.27e+06(+)	5.10e+07(+)	3.31e+06(+)
		interval	[6.00e+03, 8.20e+05]	[7.22e+05, 8.66e+06]	[2.12e+05, 5.04e+06]	[1.76e+06, 9.74e+06]	[-2.28e+03, 2.51e+05]	[-5.72e+03, 3.93e+06]	[-3.15e+05, 1.25e+06]	[3.64e+05, 3.81e+06]	[1.19e+07, 2.87e+07]	[2.47e+06, 1.41e+07]	[1.51e+07, 8.70e+07]	[-5.31e+05, 7.15e+06]
		power	-	0.98	0.95	0.99	0.03	0.90	0.31	0.93	1.00	1.00	1.00	0.94
	T2	mean	3.60e+05	4.41e+06(+)	2.81e+06(+)	5.78e+06(+)	2.47e+05(-)	1.93e+06(+)	5.24e+05(=)	1.86e+06(+)	2.13e+07(+)	8.44e+06(+)	4.89e+07(+)	4.16e+06(+)
		interval	[1.92e+02, 7.21e+05]	[8.28e+05, 7.98e+06]	[3.12e+05, 5.32e+06]	[1.49e+06, 1.01e+07]	[-3.87e+04, 5.32e+05]	[-7.15e+05, 4.57e+06]	[-2.34e+05, 1.28e+06]	[-1.91e+05, 3.90e+06]	[1.01e+07, 3.24e+07]	[2.26e+06, 1.46e+07]	[8.96e+06, 8.88e+07]	[-2.93e+05, 8.61e+06]
		power	-	0.99	0.96	0.99	0.05	0.89	0.39	0.90	1.00	1.00	1.00	0.97
Benchmark4	T1	mean	1.30e+03	1.30e+03(+)	1.30e+03(=)	1.30e+03(+)	1.30e+03(=)	1.30e+03(=)	1.30e+03(=)	1.30e+03(-)	1.30e+03(+)	1.30e+03(+)	1.30e+03(+)	1.30e+03(+)
		interval	[1.30e+03, 1.30e+03]	[1.30e+03, 1.30e+03]	[1.30e+03, 1.30e+03]	[1.30e+03, 1.30e+03]	[1.30e+03, 1.30e+03]	[1.30e+03, 1.30e+03]	[1.30e+03, 1.30e+03]	[1.30e+03, 1.30e+03]	[1.30e+03, 1.30e+03]	[1.30e+03, 1.30e+03]	[1.30e+03, 1.30e+03]	[1.30e+03, 1.30e+03]
		power	-	0.41	0.17	0.24	0.03	0.07	0.01	0.20	0.84	0.58	0.14	0.53
	T2	mean	1.30e+03	1.30e+03(+)	1.30e+03(+)	1.30e+03(+)	1.30e+03(-)	1.30e+03(=)	1.30e+03(-)	1.30e+03(=)	1.30e+03(+)	1.30e+03(+)	1.30e+03(+)	1.30e+03(+)
		interval	[1.30e+03, 1.30e+03]	[1.30e+03, 1.30e+03]	[1.30e+03, 1.30e+03]	[1.30e+03, 1.30e+03]	[1.30e+03, 1.30e+03]	[1.30e+03, 1.30e+03]	[1.30e+03, 1.30e+03]	[1.30e+03, 1.30e+03]	[1.30e+03, 1.30e+03]	[1.30e+03, 1.30e+03]	[1.30e+03, 1.30e+03]	[1.30e+03, 1.30e+03]
		power	-	0.13	0.11	0.40	0.25	0.11	0.14	0.02	0.89	0.67	0.42	0.69
Benchmark5	T1	mean	1.52e+03	1.57e+03(+)	1.52e+03(=)	1.53e+03(+)	1.58e+03(+)	1.53e+03(=)	1.52e+03(=)	1.51e+03(-)	1.54e+03(+)	1.54e+03(+)	1.53e+03(+)	1.54e+03(+)
		interval	[1.51e+03, 1.54e+03]	[1.54e+03, 1.60e+03]	[1.51e+03, 1.53e+03]	[1.53e+03, 1.53e+03]	[1.56e+03, 1.61e+03]	[1.52e+03, 1.54e+03]	[1.50e+03, 1.54e+03]	[1.51e+03, 1.52e+03]	[1.53e+03, 1.55e+03]	[1.53e+03, 1.54e+03]	[1.53e+03, 1.54e+03]	[1.53e+03, 1.54e+03]
		power	-	0.99	0.03	0.00	1.00	0.01	0.10	0.19	0.78	0.19	0.01	0.32
	T2	mean	1.52e+03	1.57e+03(+)	1.52e+03(=)	1.53e+03(+)	1.58e+03(+)	1.52e+03(=)	1.52e+03(=)	1.51e+03(-)	1.54e+03(+)	1.54e+03(+)	1.53e+03(+)	1.54e+03(+)
		interval	[1.51e+03, 1.53e+03]	[1.54e+03, 1.60e+03]	[1.51e+03, 1.53e+03]	[1.53e+03, 1.54e+03]	[1.54e+03, 1.61e+03]	[1.51e+03, 1.54e+03]	[1.51e+03, 1.54e+03]	[1.51e+03, 1.51e+03]	[1.54e+03, 1.54e+03]	[1.53e+03, 1.54e+03]	[1.53e+03, 1.54e+03]	[1.53e+03, 1.54e+03]
		power	-	0.99	0.02	0.31	0.99	0.06	0.11	0.18	1.00	0.78	0.30	0.87
Benchmark6	T1	mean	3.59e+05	2.54e+06(+)	1.47e+06(+)	2.40e+06(+)	1.23e+06(+)	1.21e+06(+)	9.96e+05(+)	1.20e+06(+)	8.75e+06(+)	2.13e+07(+)	2.27e+07(+)	1.96e+06(+)
		interval	[-8.84e+04, 8.06e+05]	[1.65e+05, 4.91e+06]	[-2.24e+05, 3.16e+06]	[8.39e+05, 3.96e+06]	[-1.27e+05, 2.59e+06]	[-3.48e+05, 2.77e+06]	[-8.08e+05, 2.80e+06]	[-1.11e+05, 2.51e+06]	[4.22e+06, 1.33e+07]	[5.66e+06, 3.70e+07]	[9.45e+06, 3.59e+07]	[-6.70e+05, 4.58e+06]
		power	-	0.94	0.81	0.98	0.76	0.75	0.70	0.75	1.00	1.00	1.00	0.87
	T2	mean	2.09e+05	2.33e+06(+)	1.55e+06(+)	2.08e+06(+)	8.77e+05(+)	7.18e+05(+)	7.83e+05(+)	9.40e+05(+)	9.23e+06(+)	2.11e+07(+)	2.09e+07(+)	1.79e+06(+)
		interval	[-1.10e+05, 5.28e+05]	[2.77e+04, 4.62e+06]	[-4.10e+04, 3.15e+06]	[4.05e+05, 3.75e+06]	[-2.96e+05, 2.05e+06]	[-1.42e+05, 1.58e+06]	[-3.16e+05, 1.88e+06]	[-3.38e+05, 2.22e+06]	[5.26e+06, 1.32e+07]	[5.67e+06, 3.65e+07]	[6.63e+06, 3.52e+07]	[-1.61e+06, 5.20e+06]
		power	-	0.96	0.92	0.97	0.77	0.70	0.73	0.79	1.00	1.00	1.00	0.90
Benchmark7	T1	mean	2.84e+03	3.40e+03(+)	2.98e+03(=)	3.86e+03(+)	3.04e+03(+)	3.17e+03(+)	3.04e+03(+)	2.87e+03(=)	4.09e+03(+)	4.25e+03(+)	4.33e+03(+)	2.95e+03(=)
		interval	[2.22e+03, 3.46e+03]	[2.65e+03, 4.15e+03]	[2.40e+03, 3.57e+03]	[3.55e+03, 4.17e+03]	[2.56e+03, 3.52e+03]	[2.45e+03, 3.90e+03]	[2.46e+03, 3.62e+03]	[2.35e+03, 3.39e+03]	[3.74e+03, 4.45e+03]	[3.75e+03, 4.74e+03]	[3.88e+03, 4.78e+03]	[2.39e+03, 3.51e+03]
		power	-	0.44	0.06	0.99	0.04	0.22	0.08	0.02	1.00	1.00	1.00	0.04
	T2	mean	2.97e+03	3.38e+03(+)	2.98e+03(=)	3.92e+03(+)	3.16e+03(+)	3.12e+03(+)	3.03e+03(=)	2.91e+03(=)	4.13e+03(+)	4.24e+03(+)	4.28e+03(+)	3.09e+03(=)
		interval	[2.27e+03, 3.66e+03]	[2.43e+03, 4.32e+03]	[2.26e+03, 3.70e+03]	[3.57e+03, 4.26e+03]	[2.57e+03, 3.75e+03]	[2.61e+03, 3.64e+03]	[2.49e+03, 3.57e+03]	[2.36e+03, 3.47e+03]	[3.79e+03, 4.47e+03]	[3.77e+03, 4.72e+03]	[3.81e+03, 4.74e+03]	[2.42e+03, 3.75e+03]
		power	-	0.29	0.06	0.93	0.05	0.02	0.01	0.02	1.00	0.99	1.00	0.05
Benchmark8	T1	mean	5.21e+02	5.21e+02(-)	5.21e+02(+)	5.21e+02(+)	5.20e+02(-)	5.20e+02(-)	5.21e+02(+)	5.21e+02(+)	5.21e+02(+)	5.21e+02(+)	5.21e+02(+)	5.21e+02(+)
		interval	[5.21e+02, 5.21e+02]	[5.20e+02, 5.21e+02]	[5.21e+02, 5.21e+02]	[5.21e+02, 5.21e+02]	[5.20e+02, 5.20e+02]	[5.20e+02, 5.21e+02]	[5.21e+02, 5.22e+02]	[5.21e+02, 5.21e+02]	[5.21e+02, 5.21e+02]	[5.21e+02, 5.21e+02]	[5.21e+02, 5.21e+02]	[5.21e+02, 5.21e+02]
		power	-	0.99	0.74	0.78	1.00	0.98	0.48	0.79	0.71	0.70	0.76	0.65
	T2	mean	5.21e+02	5.21e+02(-)	5.21e+02(+)	5.21e+02(+)	5.20e+02(-)	5.20e+02(-)	5.21e+02(+)	5.21e+02(+)	5.21e+02(+)	5.21e+02(+)	5.21e+02(+)	5.21e+02(+)
		interval	[5.21e+02, 5.21e+02]	[5.20e+02, 5.21e+02]	[5.21e+02, 5.21e+02]	[5.21e+02, 5.21e+02]	[5.20e+02, 5.20e+02]	[5.20e+02, 5.21e+02]	[5.21e+02, 5.21e+02]	[5.21e+02, 5.21e+02]	[5.21e+02, 5.21e+02]	[5.21e+02, 5.21e+02]	[5.21e+02, 5.21e+02]	[5.21e+02, 5.21e+02]
		power	-	0.98	0.46	0.40	1.00	0.98	0.34	0.40	0.47	0.53	0.41	0.51
Benchmark9	T1	mean	8.49e+03	8.39e+03(=)	1.28e+04(+)	1.50e+04(+)	8.26e+03(=)	7.47e+03(-)	7.51e+03(-)	7.41e+03(-)	1.51e+04(+)	1.49e+04(+)	1.50e+04(+)	8.98e+03(=)
		interval	[6.20e+03, 1.08e+04]	[6.75e+03, 1.00e+04]	[7.18e+03, 1.85e+04]	[1.40e+04, 1.59e+04]	[6.50e+03, 1.00e+04]	[5.80e+03, 9.14e+03]	[6.39e+03, 8.63e+03]	[5.44e+03, 9.38e+03]	[1.45e+04, 1.56e+04]	[1.41e+04, 1.58e+04]	[1.41e+04, 1.58e+04]	[4.71e+03, 1.32e+04]
		power	-	0.01	0.77	1.00	0.01	0.07	0.01	0.11	1.00	1.00	1.00	0.30
	T2	mean	1.62e+03	1.62e+03(+)	1.62e+03(+)	1.62e+03(+)	1.62e+03(-)	1.62e+03(=)	1.62e+03(=)	1.62e+03(-)	1.62e+03(+)	1.62e+03(+)	1.62e+03(+)	1.62e+03(+)
		interval	[1.62e+03, 1.62e+03]	[1.62e+03, 1.62e+03]	[1.62e+03, 1.62e+03]	[1.62e+03, 1.62e+03]	[1.62e+03, 1.62e+03]	[1.62e+03, 1.62e+03]	[1.62e+03, 1.62e+03]	[1.62e+03, 1.62e+03]	[1.62e+03, 1.62e+03]	[1.62e+03, 1.62e+03]	[1.62e+03, 1.62e+03]	[1.

TABLE S.III
COMPLETE EXPERIMENTAL RESULTS OF AEDMTO AND COMPARED ALGORITHMS ON SCP APPLICATION

SCP			AEDMTO	AEMTO	MaTDE	BLKT-DE	TRADE
Case 1	T1	mean	7.11e+01	1.06e+02(+)	1.63e+02(+)	9.64e+01(+)	1.56e+02(+)
		interval	[59.32,82.97]	[83.29,129.06]	[62.04,264.79]	[25.24,167.56]	[59.17,252.58]
		power	-	0.98	0.96	0.80	0.96
	T2	mean	6.53e+01	9.77e+01(+)	1.54e+02(+)	8.89e+01(+)	1.41e+02(+)
		interval	[55.21,75.32]	[77.51,117.84]	[50.21,257.31]	[32.76,144.96]	[52.49,229.35]
		power	-	0.99	0.96	0.80	0.96
Number of "+/=-/-"			2/0/0	2/0/0	2/0/0	2/0/0	
Case 2	T1	mean	7.29e+01	1.64e+02(+)	1.98e+02(+)	1.75e+02(+)	1.77e+02(+)
		interval	[55.30,90.44]	[115.07,213.65]	[102.92,293.35]	[77.34,273.31]	[97.56,256.01]
		power	-	1.00	0.99	0.96	0.98
	T2	mean	6.56e+01	1.52e+02(+)	1.86e+02(+)	1.69e+02(+)	1.58e+02(+)
		interval	[54.70,76.48]	[95.60,208.89]	[84.42,287.40]	[76.25,262.01]	[86.88,229.59]
		power	-	1.00	0.99	0.98	0.99
	T3	mean	6.03e+01	1.41e+02(+)	1.77e+02(+)	1.63e+02(+)	1.44e+02(+)
		interval	[50.26,70.36]	[86.12,194.95]	[81.72,272.75]	[69.10,257.10]	[75.57,213.24]
		power	-	0.99	0.99	0.98	0.99
	T4	mean	5.71e+01	1.35e+02(+)	1.74e+02(+)	1.51e+02(+)	1.39e+02(+)
		interval	[45.94,68.30]	[80.72,190.13]	[76.25,271.21]	[55.20,246.93]	[65.75,211.65]
		power	-	0.99	0.99	0.97	0.98
	T5	mean	5.64e+01	1.33e+02(+)	1.66e+02(+)	1.53e+02(+)	1.36e+02(+)
		interval	[47.54,65.23]	[80.47,184.67]	[71.23,260.37]	[66.97,238.09]	[54.25,218.68]
		power	-	1.00	0.99	0.99	0.97
	T6	mean	5.50e+01	1.32e+02(+)	1.60e+02(+)	1.41e+02(+)	1.19e+02(+)
		interval	[45.93,63.99]	[77.48,186.23]	[75.34,244.49]	[49.47,232.39]	[45.36,192.68]
		power	-	0.99	0.99	0.97	0.95
Number of "+/=-/-"			6/0/0	6/0/0	6/0/0	6/0/0	
Case 3	T1	mean	7.06e+01	2.04e+02(+)	2.18e+02(+)	1.64e+02(+)	1.62e+02(+)
		interval	[59.98,81.30]	[185.47,222.91]	[130.96,305.78]	[50.37,276.78]	[70.73,253.84]
		power	-	1.00	1.00	0.96	0.97
	T2	mean	6.52e+01	1.94e+02(+)	2.00e+02(+)	1.61e+02(+)	1.58e+02(+)
		interval	[55.13,75.18]	[171.49,217.10]	[118.92,280.66]	[41.56,280.89]	[85.78,230.32]
		power	-	1.00	1.00	0.96	0.99
	T3	mean	6.08e+01	1.80e+02(+)	1.82e+02(+)	1.51e+02(+)	1.45e+02(+)
		interval	[49.62,71.91]	[156.97,202.74]	[94.54,269.06]	[44.31,257.21]	[65.41,224.67]
		power	-	1.00	0.99	0.96	0.97
	T4	mean	5.62e+01	1.69e+02(+)	1.81e+02(+)	1.42e+02(+)	1.28e+02(+)
		interval	[48.61,63.86]	[147.76,189.84]	[87.61,275.23]	[44.65,240.21]	[36.58,219.42]
		power	-	1.00	1.00	0.97	0.96
	T5	mean	5.51e+01	1.66e+02(+)	1.70e+02(+)	1.39e+02(+)	1.23e+02(+)
		interval	[47.58,62.62]	[144.08,188.68]	[79.22,260.38]	[44.95,233.50]	[44.29,202.48]
		power	-	1.00	0.99	0.97	0.96
	T6	mean	5.41e+01	1.62e+02(+)	1.64e+02(+)	1.31e+02(+)	1.29e+02(+)
		interval	[47.24,60.90]	[144.45,178.81]	[59.53,268.50]	[36.11,225.45]	[42.42,216.38]
		power	-	1.00	0.99	0.97	0.97
	T7	mean	5.47e+01	1.54e+02(+)	1.55e+02(+)	1.32e+02(+)	1.27e+02(+)
		interval	[46.78,62.66]	[138.11,170.08]	[61.48,248.30]	[40.31,224.12]	[48.30,205.14]
		power	-	1.00	0.99	0.97	0.97
	T8	mean	5.42e+01	1.52e+02(+)	1.50e+02(+)	1.30e+02(+)	1.12e+02(+)
		interval	[48.13,60.19]	[136.78,168.16]	[59.61,240.64]	[43.75,215.61]	[33.71,190.34]
		power	-	1.00	0.99	0.97	0.96
	T9	mean	5.39e+01	1.49e+02(+)	1.43e+02(+)	1.28e+02(+)	1.22e+02(+)
		interval	[49.34,58.42]	[136.86,161.99]	[63.00,223.28]	[51.32,204.22]	[43.79,199.24]
		power	-	1.00	0.99	0.98	0.98
	T10	mean	5.47e+01	1.44e+02(+)	1.39e+02(+)	1.20e+02(+)	1.22e+02(+)
		interval	[51.60,57.71]	[126.04,161.36]	[58.47,219.75]	[45.66,194.93]	[54.24,188.97]
		power	-	1.00	0.99	0.99	0.99
	T11	mean	5.52e+01	1.44e+02(+)	1.40e+02(+)	1.21e+02(+)	1.07e+02(+)
		interval	[50.49,59.87]	[130.51,157.46]	[61.03,219.46]	[47.11,194.20]	[22.37,191.87]
		power	-	1.00	0.99	0.98	0.96
Number of "+/=-/-"			11/0/0	11/0/0	11/0/0	11/0/0	
Summary			19/0/0	19/0/0	19/0/0	19/0/0	

S.III. CONVERGENCE CURVES OF AEDMTO AND COMPARED EMTO ALGORITHMS

A. Convergence Curves of AEDMTO and Compared Algorithms on CEC2017

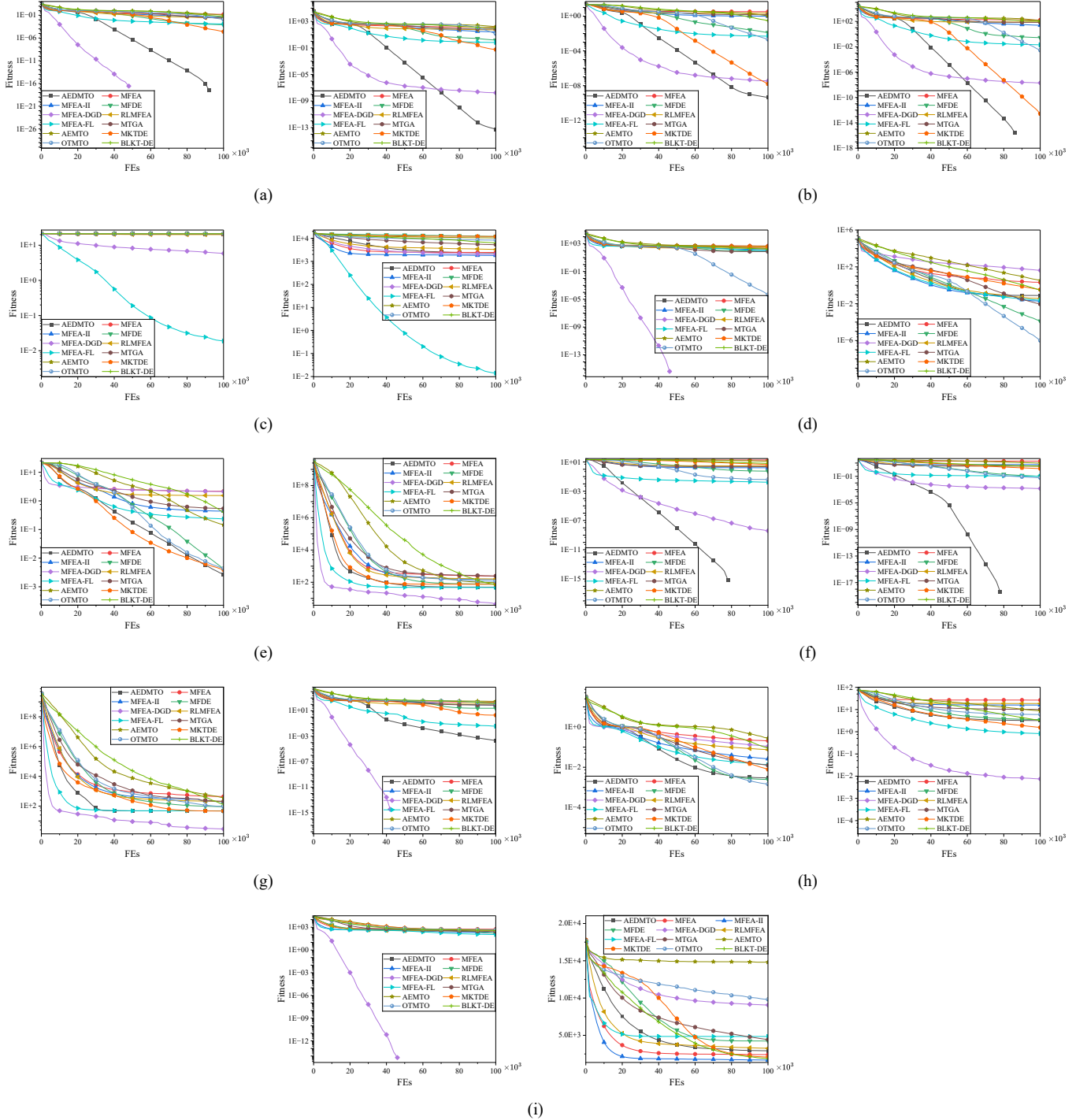


Fig. S1. Convergence curves of AEDMTO and compared EMTO algorithms on all tasks in CEC2017. (a) CIHS; (b) CIMS; (c) CILS; (d) PIHS; (e) PIMS; (f) PILS; (g) NIHS; (h) NIMS; (i) NILS.

B. Convergence Curves of AEDMTO and Compared Algorithms on CEC2022

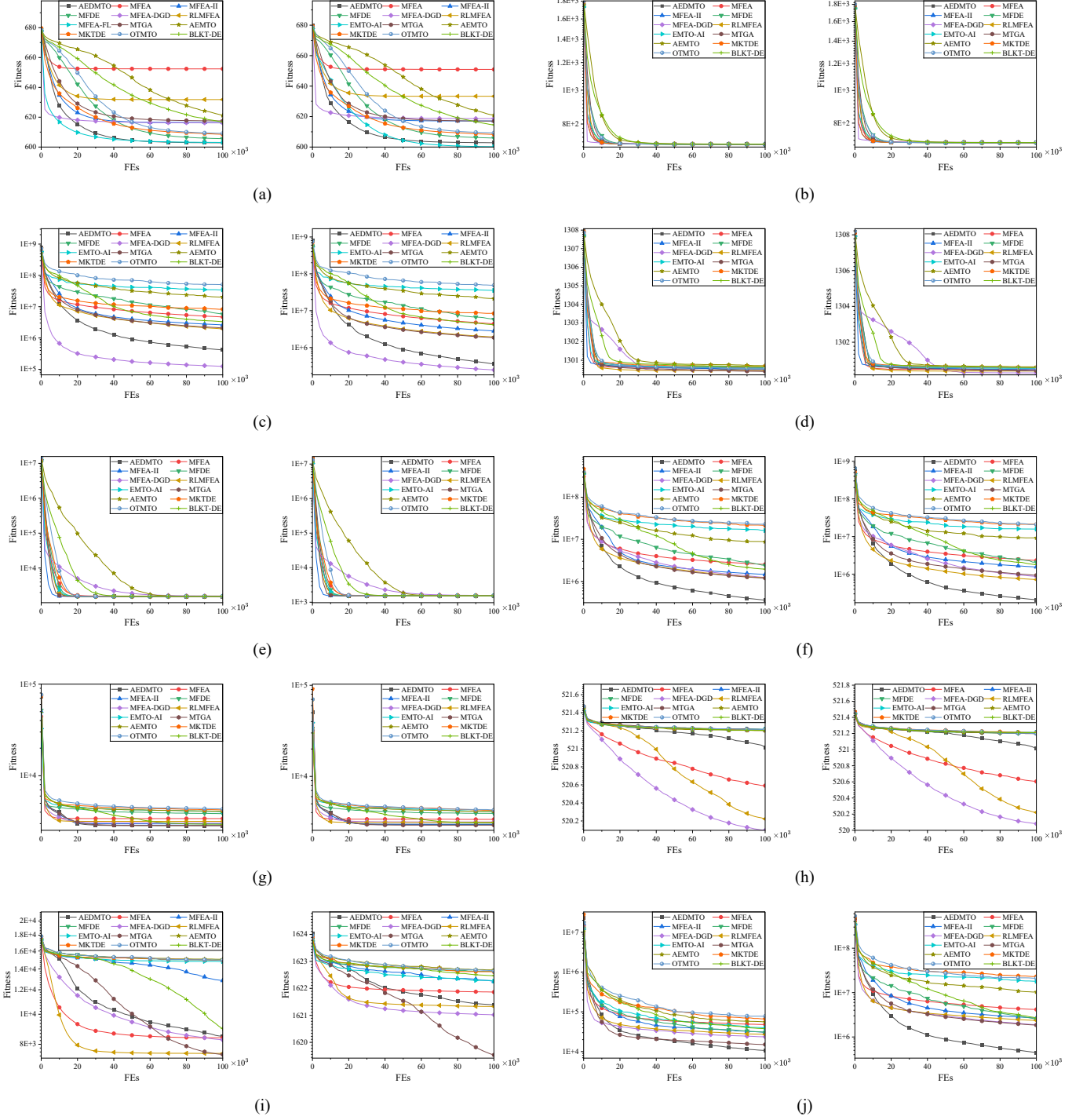


Fig. S2. Convergence curves of AEDMTO and compared EMTO algorithms on all tasks in CEC2022. Subplots (a)-(j) correspond to the ten benchmarks, respectively.

S.IV. EXTENSIVE ABLATION STUDIES

A. Justification of Gaussian Assumption

In AEDMTO, the unimodal Gaussian distribution may not always be theoretically guaranteed across all optimization landscapes, such as multimodal, heavier-tailed, skewed, or uniform. Without prior knowledge of the landscapes of problems, we adopt a simple unimodal Gaussian model to describe the distribution of tasks due to its simplicity and widespread use [S3]-[S5]. To justify the unimodal Gaussian model, we generate four AEDMTO variants with multimodal Gaussian distribution (Gaussian mixture distribution)-based AED, Cauchy distribution (one of the heavier-tailed distribution models)-based AED, skewed distribution-based AED, and uniform distribution-based AED, denoted as AEDMTO-w/-multimodal, AEDMTO-w/-Cauchy, AEDMTO-w/-skew, and AEDMTO-w/-uniform, respectively.

The multimodal Gaussian distribution is the mixture distribution of multiple Gaussian probability distribution models [S6]. In AEDMTO-w/-multimodal, we employ K (set as 5) Gaussian distribution components to describe the distribution of each dimension in the target task. First, the K -means is executed to split individuals into K clusters. Note that to avoid the ‘‘curse of dimensionality’’ [S7], individuals will be reduced to 3 dimensions by principal component analysis when clustered. Then, the centroid and the standard deviation of each cluster are calculated. By combining the parameters of each Gaussian component, we can obtain the new distribution description, shown as Eq. (S17):

$$ds_j(\mathbf{XS}) = \sum_{k=1}^K \frac{N_k}{N} \exp\left(\frac{-(XS_j - \text{cntr}_{k,j})^2}{2 \times (\text{std}_{k,j})^2}\right) \quad (\text{S17})$$

where $\text{cntr}_{k,j}$ and $\text{std}_{k,j}$ are the centroid and standard deviation of j^{th} dimension in the k^{th} cluster, respectively, N_k denotes the number of individuals in the k^{th} cluster.

The Cauchy distribution is one of the heavier-tailed distributions, with tails heavier than those of the Gaussian distribution, i.e., extreme values occur with a higher probability than in the Gaussian distribution. We have tried to use the Cauchy distribution [S8] in AED strategy to estimate the distribution of each dimension in the target task using Eq. (S18):

$$ds_j(\mathbf{XS}) = \frac{1}{1 + \left(\frac{x - \text{cntr}_j}{\text{std}_j}\right)^2} \quad (\text{S18})$$

In AEDMTO-w/-skew, we assume that each dimension follows a skewed distribution with a bias between the mean and median. The distribution for each dimension is shown in Eq. (S19):

$$ds_j(\mathbf{XS}) = \frac{2}{\text{std}_j} \times \varphi\left(\frac{XS_j - \text{cntr}_j}{\text{std}_j}\right) \times \Phi\left(SK \times \frac{XS_j - \text{cntr}_j}{\text{std}_j}\right) \quad (\text{S19})$$

$$SK = \frac{1}{N} \sum_{i=1}^N \frac{(XT_i^j - \text{cntr}_j)^3}{\text{std}_j^3} \quad (\text{S20})$$

where $\varphi(*)$ and $\Phi(*)$ are the probability density function and distribution function of the standard Gaussian distribution $N(0, 1)$, SK is the skewness of the skewed distribution.

In AEDMTO-w/-uniform, for each dimension, we first search for the minimum and maximum values for each dimension across all individuals, as:

$$lb_j = \min\{XT_i^j \mid i = 1, 2, \dots, N\} \quad (\text{S21})$$

$$hb_j = \max\{XT_i^j \mid i = 1, 2, \dots, N\} \quad (\text{S22})$$

Then, for each individual in the source task, the similarity of the j^{th} dimension to the target task is described by Eq. (S23):

$$ds_j(\mathbf{XS}) = \begin{cases} 1, & \text{if } XS_j \in [lb_j, hb_j] \\ 0, & \text{otherwise} \end{cases} \quad (\text{S23})$$

We have compared AEDMTO with these four variants and the experimental results are shown in Table S.IV. From Table S.IV, we find that AEDMTO and its variants achieve similar performance on at least 13 of 18 tasks, indicating that AEDMTO is relatively insensitive to different distribution models. Note that the core idea of AED is to measure similarity from both individual and population aspects, rather than relying on a specific distribution model. Therefore, we employ the simple unimodal Gaussian distribution-based AED in AEDMTO.

TABLE S.IV
THE COMPARISON AMONG AEDMTO WITH DIFFERENT DISTRIBUTION MODELS

CEC2017		AEDMTO	AEDMTO-w/-multimodal	AEDMTO-w/-Cauchy	AEDMTO-w/-skew	AEDMTO-w/-uniform
CIHS	T1	0.00e+00	0.00e+00(=)	0.00e+00(=)	0.00e+00(=)	0.00e+00(=)
	T2	5.33e-14	6.66e-14(+)	1.95e-13(+)	3.45e-14(=)	8.64e-14(+)
CIMS	T1	4.40e-10	4.82e-10(=)	7.45e-10(+)	3.86e-10(=)	4.86e-10(=)
	T2	0.00e+00	0.00e+00(=)	0.00e+00(=)	0.00e+00(=)	0.00e+00(=)
CILS	T1	2.08e+01	2.08e+01(=)	2.08e+01(=)	2.01e+01(=)	2.08e+01(=)
	T2	2.33e+03	2.75e+03(+)	2.94e+03(+)	2.31e+03(=)	2.66e+03(=)
PIHS	T1	2.07e+02	1.70e+02(=)	1.92e+02(=)	2.79e+02(+)	1.78e+02(=)
	T2	7.79e-02	2.20e-04(=)	9.63e-02(=)	1.68e-01(+)	1.70e-03(=)
PIMS	T1	2.74e-03	3.53e-03(=)	4.66e-03(-)	2.09e-02(+)	1.94e-03(-)
	T2	4.99e+01	4.65e+01(=)	4.64e+01(=)	5.31e+01(=)	4.99e+01(=)
PILS	T1	0.00e+00	0.00e+00(=)	0.00e+00(=)	0.00e+00(=)	0.00e+00(=)
	T2	0.00e+00	0.00e+00(=)	0.00e+00(=)	0.00e+00(=)	0.00e+00(=)
NIHS	T1	4.79e+01	4.80e+01(=)	4.79e+01(=)	4.80e+01(=)	4.79e+01(=)
	T2	2.09e-04	1.68e-04(=)	2.21e-06(-)	6.34e-05(-)	7.11e-06(-)
NIMS	T1	2.97e-03	1.64e-03(=)	7.26e-04(=)	2.93e-03(+)	2.68e-03(=)
	T2	3.19e+00	3.35e+00(=)	3.37e+00(=)	3.45e+00(=)	3.16e+00(=)
NILS	T1	1.68e+02	1.63e+02(=)	1.50e+02(=)	1.94e+02(=)	1.97e+02(+)
	T2	2.93e+03	3.01e+03(=)	3.05e+03(=)	2.93e+03(=)	3.09e+03(=)
Number of “+/-/-”		2/16/0		3/13/2	4/13/1	2/14/2

B. Justification of Exponential Weighting

We define the KT frequency f_{KT} as the weighted average of the ds for all individuals from the source task as:

$$f_{KT} = \sum_{i=1}^N \frac{1}{2^i} \times ins(\mathbf{S}_i) \quad (S24)$$

where $\mathbf{S}_i$ is the i^{th} individual sorted in descending order by its ins values (the similarity measurement between the i^{th} individual in the source task and the entire target task), and the weight $1 / 2^i$ is positively correlated with the ins values. Therefore, the core idea of exponential weighting is to assign higher weights to individuals with higher similarities, rather than to specify a particular weighting method. Therefore, using different weighting methods does not affect this core idea and may all work well, though slight performance differences may exist. Herein, we first prove the rationality of the exponential weights, which ensures $0 \leq f_{KT} \leq 1$.

Proof: The sequence of the exponential weights $1 / 2^i$ is a geometric sequence, whose sum is close to or a little less than 1. Meanwhile, the similarity $ins(\mathbf{XS})$ between an individual $\mathbf{XS}$ and the entire target task is defined as the average ds of all dimensions in $\mathbf{XS}$, shown in Eq. (5) of the manuscript, also given in Eq. (S25) herein:

$$ins(\mathbf{XS}) = \frac{1}{D_t} \sum_{j=1}^{D_t} ds_j(\mathbf{XS}_M) \quad (S25)$$

and the similarity $ds_j(\mathbf{XS})$ between the j^{th} dimension XS_j of individual $\mathbf{XS}$ in the source task and the j^{th} dimension of the entire target task is calculated as Eq. (4) of the manuscript, also given in Eq. (S26) herein:

$$ds_j(\mathbf{XS}) = \exp \left(-\frac{(XS_j - cntr_j)^2}{2 \times (std_j)^2} \right) \quad (S26)$$

where $cntr_j$ and std_j stand for the centroid and standard deviation of the j^{th} dimension for all individuals in the target task, respectively. From Eq. (S26), we find that for any dimension of $\mathbf{XS}$, $ds_j(\mathbf{XS})$ satisfies $0 \leq ds_j(\mathbf{XS}) \leq 1$, while the $ins(\mathbf{XS})$ of each individual $\mathbf{XS}$ in Eq. (S25), as the average of $ds_j(\mathbf{XS})$, also satisfies $0 \leq ins(\mathbf{XS}) \leq 1$. Hence, the KT frequency f_{KT} , as the weighted average of $ins(\mathbf{XS})$ ($ins(\mathbf{XS}) \in [0, 1]$), the sum of weights is close to or a little less than 1), also satisfies $0 \leq f_{KT} \leq 1$.

To explore the suitable weighting mechanism, we generate AEDMTO variants using other weighting methods, including linear weighting and logarithm weighting, denoted as AEDMTO-w/-linear and AEDMTO-w/-log, respectively. With the linear weighting and logarithm weighting, the KT frequency f_{KT} is calculated as Eqs. (S27) and (S28):

$$f_{KT} = \sum_{i=1}^N \frac{2 \times i}{N \times (N + 1)} \times ins(\mathbf{S}_i) \quad (S27)$$

$$f_{KT} = \sum_{i=1}^N \frac{\log(i)}{\log(N!)} \times ins(\mathbf{S}_i) \quad (S28)$$

The results are shown in Table S.V. From Table S.V, we find that AEDMTO and its variants obtain similar performance on 17 of 18 tasks, which indicates that AEDMTO is relatively insensitive to different weighting models. Therefore, the exponential weighting is finally employed in AED strategy.

TABLE S.V
THE COMPARISON AMONG AEDMTO, AEDMTO-W/-LINEAR, AND AEDMTO-W/-LOG

CEC2017		AEDMTO	AEDMTO-w/-linear	AEDMTO-w/-log
CIHS	T1	0.00e+00	0.00e+00(=)	0.00e+00(=)
	T2	5.33e-14	6.34e-14(=)	6.90e-14(=)
CIMS	T1	4.40e-10	4.66e-10(=)	5.04e-10(+)
	T2	0.00e+00	0.00e+00(=)	0.00e+00(=)
CILS	T1	2.08e+01	2.08e+01(=)	2.08e+01(=)
	T2	2.33e+03	2.83e+03(+)	2.55e+03(=)
PIHS	T1	2.07e+02	1.62e+02(=)	1.70e+02(=)
	T2	7.79e-02	7.68e-05(=)	1.06e-02(=)
PIMS	T1	2.74e-03	2.12e-03(=)	8.83e-03(=)
	T2	4.99e+01	5.03e+01(=)	4.67e+01(=)
PILS	T1	0.00e+00	0.00e+00(=)	3.25e-05(=)
	T2	0.00e+00	0.00e+00(=)	1.84e-01(=)
NIHS	T1	4.79e+01	4.79e+01(=)	4.80e+01(=)
	T2	2.09e-04	3.68e-01(=)	1.65e-04(=)
NIMS	T1	2.97e-03	3.19e-03(=)	2.68e-03(=)
	T2	3.19e+00	3.70e+00(=)	3.46e+00(=)
NILS	T1	1.68e+02	1.68e+02(=)	1.64e+02(=)
	T2	2.93e+03	3.15e+03(=)	3.12e+03(=)
Number of “+/-/-”			1/17/0	1/17/0

C. Scalability of AEDMTO on High-Dimensional Scenarios

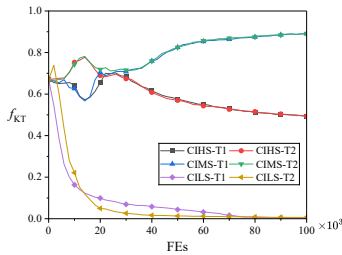
In this section, AEDMTO is extended to high-dimensional scenarios by enlarging the dimensionalities of tasks in CEC2017 by 20 times. On high-dimensional scenarios, we first compare AEDMTO with its variants with fixed KT frequency values of 0.0, 0.1, 0.3, 0.5, 0.7, 0.8, and 1.0. The comparison results are shown in Table S.VI. From Table S.VI, we find that AEDMTO still outperforms all its variants with fixed KT frequency on high-dimensional MTOPs. Specifically, AEDMTO dominates its variants with fixed f_{KT} values on at least 8 tasks, while none of the variants can surpass AEDMTO on more than 2 tasks, which fully shows the scalability of AED strategy on high-dimensional scenarios.

Next, we continue to plot the variation curves of the f_{KT} values on these high-dimensional MTOPs in Fig. S3. From Fig. S3, we find that in CI group, tasks in CIHS and CIMS adjust their KT frequency to higher values, while the KT frequency for tasks in CILS decreases gradually to lower values to avoid negative KT. Similarly, in NI group, tasks in NIHS maintain a higher KT frequency during evolutionary process, while tasks in NIMS and NILS decrease the KT frequency. Note that in PI group, the variation curves seem to be inconsistent with the problem definition of CEC2017, where tasks in PILS maintain a higher KT frequency, while tasks in PIMS and PIHS obtain a medium and a lower KT frequency. However, according to the experimental results in Table S.VI, the curves still show consistent trends with the experimental results, where tasks in PIHS obtain better results with a lower KT frequency, tasks in PIMS and PILS perform better with medium and higher KT frequency.

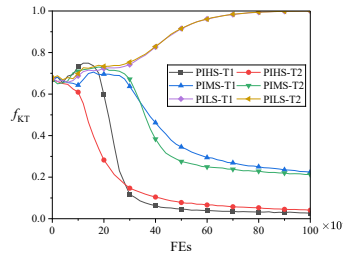
The experimental results show that even on high-dimensional scenarios, AED can still sample the suitable mapping and the adaptive KT frequency mechanism in AEDMTO is still effective. For tasks with high similarity, suitable dimension-matching schemes are easier to sample, in that case, f_{KT} will rise to a high value to allow more high-quality information to be transferred, even when tasks have higher dimensionalities. Meanwhile, for tasks with low similarity, suitable mapping may be harder to sample, at this time, f_{KT} decreases with the evolutionary process, which filters the unsuitable dimension-matching schemes. This phenomenon fully illustrates the scalability of AEDMTO on high-dimensional scenarios.

TABLE S.VI
THE COMPARISON AMONG AEDMTO AND ITS VARIANTS WITH FIXED KT FREQUENCY ON HIGH-DIMENSIONAL SCENARIOS

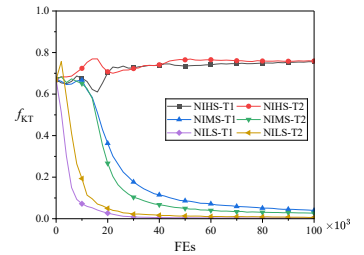
CEC2017	Adaptive f_{KT}	$f_{KT} = 0.0$	$f_{KT} = 0.1$	$f_{KT} = 0.3$	$f_{KT} = 0.5$	$f_{KT} = 0.7$	$f_{KT} = 0.8$	$f_{KT} = 1.0$
CIHS	T1	1.75e-02	3.82e+00(+)	5.58e-01(+)	2.72e-02(=)	3.10e-02(=)	2.06e-02(=)	3.82e-02(=)
	T2	3.34e+02	1.23e+04(+)	5.12e+03(+)	4.98e+02(=)	5.60e+02(=)	4.01e+02(=)	7.43e+02(=)
CIMS	T1	7.85e-02	8.46e+00(+)	2.18e+00(+)	2.71e-01(+)	1.54e-01(=)	1.57e-01(=)	6.99e-02(=)
	T2	7.54e+01	1.23e+04(+)	4.22e+03(+)	4.75e+02(+)	2.50e+02(=)	2.65e+02(=)	5.64e+01(=)
CILS	T1	1.54e+01	1.58e+01(+)	1.57e+01(+)	1.58e+01(+)	1.58e+01(+)	1.59e+01(+)	1.60e+01(+)
	T2	2.67e+05	2.58e+05(-)	2.64e+05(=)	2.76e+05(+)	2.85e+05(+)	2.90e+05(+)	2.92e+05(+)
PIHS	T1	5.08e+03	1.21e+04(+)	5.48e+03(+)	5.53e+03(+)	5.37e+03(+)	5.71e+03(+)	5.47e+03(+)
	T2	8.14e+04	4.20e+05(+)	9.27e+04(+)	9.12e+04(+)	9.30e+04(+)	9.52e+04(+)	1.00e+05(+)
PIMS	T1	6.93e-01	8.60e+00(+)	9.07e-01(+)	7.43e-01(=)	8.00e-01(=)	8.13e-01(+)	8.20e-01(+)
	T2	6.00e+03	3.09e+09(+)	1.77e+04(+)	8.62e+03(+)	9.11e+03(=)	6.20e+03(=)	5.16e+03(=)
PILS	T1	1.72e-03	8.53e+00(+)	1.46e+00(+)	8.42e-02(+)	4.01e-02(+)	4.06e-03(+)	2.45e-03(=)
	T2	8.90e-01	1.79e+02(+)	3.70e+01(+)	7.12e+00(+)	5.15e+00(+)	1.59e+00(+)	1.19e+00(=)
NIHS	T1	2.16e+03	3.20e+09(+)	2.15e+06(+)	3.07e+03(+)	2.59e+03(+)	6.39e+03(=)	2.45e+03(=)
	T2	2.29e+02	1.23e+04(+)	3.70e+03(+)	2.30e+02(=)	2.60e+02(+)	5.11e+02(=)	2.48e+02(=)
NIMS	T1	1.23e+00	4.02e+00(+)	1.57e+00(+)	1.33e+00(+)	1.38e+00(+)	1.38e+00(+)	1.35e+00(+)
	T2	1.03e+02	4.18e+02(+)	1.83e+02(+)	1.20e+02(+)	1.05e+02(=)	1.06e+02(=)	1.09e+02(=)
NILS	T1	1.24e+04	1.25e+04(=)	1.25e+04(=)	1.29e+04(=)	1.32e+04(+)	1.33e+04(+)	1.33e+04(+)
	T2	2.67e+05	2.58e+05(-)	2.67e+05(=)	2.76e+05(+)	2.83e+05(+)	2.92e+05(+)	2.96e+05(+)
Number of “+/-”		15/1/2	15/3/0	13/5/0	11/7/0	10/8/0	8/10/0	8/10/0



(a)



(b)



(c)

Fig. S3. The KT frequency variation curves for the three groups on high-dimensional scenarios. (a) CI group; (b) PI group; (c) NI group.

D. Analysis of Dimension Selection in SDM

In SDM strategy, the least-repeated selection is employed to fully exploit the information from all dimensions in source task. To illustrate the effectiveness of this restriction, AEDMTO is further compared with its variant with repeated selection of dimensions in SDM, denoted as AEDMTO-SDM-RS, where the dimension-matching is completely random. The detailed comparison results between AEDMTO and AEDMTO-SDM-RS on CEC2022 are shown in Table S.VII.

From Table S.VII, we find that AEDMTO outperforms AEDMTO-SDM-RS on 5 tasks, while AEDMTO-SDM-RS fails to show superiority on any task. This phenomenon fully shows that with the non-repeated selection of dimensions, the potential useful index-unaligned dimensions are more likely to match with each other, which enables sufficient knowledge exchange between index-unaligned dimensions and preserves the diversity of dimension-matching schemes.

TABLE S.VII
THE COMPARISON BETWEEN AEDMTO AND AEDMTO-SDM-RS

CEC2022		AEDMTO	AEDMTO-SDM-RS
Benchmark1	T1	6.03e+02	6.03e+02(+)
	T2	6.03e+02	6.03e+02(+)
Benchmark2	T1	7.00e+02	7.00e+02(=)
	T2	7.00e+02	7.00e+02(+)
Benchmark3	T1	4.13e+05	4.64e+05(=)
	T2	3.60e+05	5.27e+05(+)
Benchmark4	T1	1.30e+03	1.30e+03(=)
	T2	1.30e+03	1.30e+03(=)
Benchmark5	T1	1.52e+03	1.52e+03(=)
	T2	1.52e+03	1.52e+03(=)
Benchmark6	T1	3.59e+05	3.85e+05(=)
	T2	2.09e+05	3.02e+05(+)
Benchmark7	T1	2.84e+03	2.88e+03(=)
	T2	2.97e+03	2.89e+03(=)
Benchmark8	T1	5.21e+02	5.21e+02(=)
	T2	5.21e+02	5.21e+02(=)
Benchmark9	T1	8.49e+03	8.72e+03(=)
	T2	1.62e+03	1.62e+03(=)
Benchmark10	T1	1.06e+04	1.11e+04(=)
	T2	4.46e+05	4.47e+05(=)
Number of “+/-/-”		5/15/0	

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